Evaluating AI Competency Development in Higher Education: A Multi-Method Survey Report

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Contents

1. Introduction

This report documents my exploratory evaluation of an AI Competency instrument used before and after the 2025 AI Institute for Teaching & Learning. I began by examining item distributions, ordinal correlations, reliability, dimensionality, and redundancy because the survey was newly developed and its measurement properties were not yet established.

The analysis records both useful and unsuccessful routes. I explored factor and bifactor structures, examined whether highly related items could be pruned or bundled, and documented where the small sample and response patterns made more complex models unstable. For the public pre/post summary, I return to the five competency areas specified in the final survey instruments and interpret the results as exploratory changes in self-reported competency, not as instrument validation or causal evidence of Institute effectiveness.

2. AI Competency Questions

Participants ($n = 47$) completed the 17-item AI Competency questions before (pre) and after (post) a three-day AI Institute. Items were rated on a 5-point ordered scale $x_{ij} \in \{1, \dots, 5\}$, $1 =$ No awareness; $2 =$ Awareness (Recognizes AI exists but has limited knowledge and minimal experience using it in teaching); $3 =$ Acquire (Basic understanding of AI and its applications in teaching); $4 =$ Deepen (Skilled integration of AI into pedagogical practices); and $5 =$ Create (Advanced ability to customize and innovate in teaching). The AI Competency questions are designed to assess five competency areas:

1. **Human-Centered Mindset**: Understanding human–AI dynamics and protecting human agency.
2. **Ethics of AI**: Identifying ethical concerns and understanding potential harms.
3. **AI Foundations and Applications**: Evaluating AI tools' capabilities, limitations, and appropriate use.
4. **AI Pedagogy**: Designing learning activities and facilitating student use of AI.
5. **AI for Professional Development**: Using AI for lifelong learning and professional growth.

This report (1) assesses dimensionality, redundancy, and reliability, (2) documents trial score-reduction and attempted confirmatory routes, and (3) summarizes paired pre/post change.

2.1 Sources of Covariance

1. General AI Competence – a broad skill that elevates scores across all items.
2. Specific Competency Factors – targeted constructs within each of the five domains.
3. Item-Level Redundancy – wording overlap that inflates correlations beyond 1 & 2.

To investigate these sources, I examine exploratory factor and bifactor solutions, inspect item overlap, and conduct a trial pruning exercise.

I provide the step-by-step psychometric workflow used to assess redundancy, reliability, and the feasibility of more complex measurement models.

2.2 Data & Sample

Our intent was to survey all attendees of the AI Institute. Sixty-nine faculty completed a participation form to attend the AI Institute, and we used those forms to create the sample frame. All those who completed a participation form and who were not on the planning or research team were sent a pre-survey. From this solicitation, six attendees opted out of our study. For the pre-survey, three did not respond, and three opened but did not complete the survey. These twelve participants did not receive a post-survey solicitation. Of those who received a post-survey solicitation, four people *did not respond*. No one opted out of the post-survey. All participants *who responded* to the post-survey completed it. We removed the following participant rows from the `institute_data` dataset:

1. Those who opted out ($n = 6$)
2. Those who completed the participation form but were also on the AI Institute planning or research team ($n = 3$)
3. Those who informed the research team that they would not be able to attend the AI Institute ($n = 3$)
4. Those who did not respond or complete the pre-survey ($n = 6$)
5. Those who did not respond to the post-survey ($n = 4$)
6. Those who attended the AI Institute but did not complete the participation form and therefore did not receive any survey solicitations. ($n =$unknown)

- File: `institute_data from probability_data.rds`
- Rows: 47 complete cases (post-filtering)
- Variables: 17 items × 2 waves (_pre, _post suffixes)
- Design: Within-subjects (pre- vs. post-survey)

Small-N note: With N = 47 and 17 ordinal items, we sit near the lower bound for stable factor estimation. I therefore treat the factor analyses as exploratory diagnostics and do not assume that more complex confirmatory or longitudinal models will be estimable.

2.3 Global (All-Items) Analysis

We begin by looking at the overall item means and variances for the pre-survey. This will help us understand the baseline distribution of responses before the Institute. A mean close to the center of the range of all possible scores is desirable, as it indicates that respondents are not clustered at the extremes of the scale.

Table 2.1: Item means and variances (pre-survey, N = 47)

item	mean	var
basic_capable_limits_pre	2.85	0.74
collaboration_pre	2.62	0.98
creative_uses_pre	2.15	1.13
data_privacy_intellectual_property_pre	2.32	1.05
decision_to_use_pre	2.38	1.07
design_learning_activities_pre	2.38	0.72
ethical_concerns_in_applications_pre	2.83	0.80
facilitate_student_use_to_learn_pre	2.38	0.68
human_accountability_pre	2.57	0.90
humans_versus_ai_control_pre	2.64	0.80
pre_use_boundaries_goals_constraints_pre	2.91	1.25
professional_development_pre	2.53	0.86
protect_human_agency_pre	2.36	0.76
societal_impact_citizenship_pre	2.72	0.68
specific_use_case_pre	2.34	0.97
systems_cause_harm_pre	2.89	0.66
when_to_pause_mitigate_pre	2.23	0.88

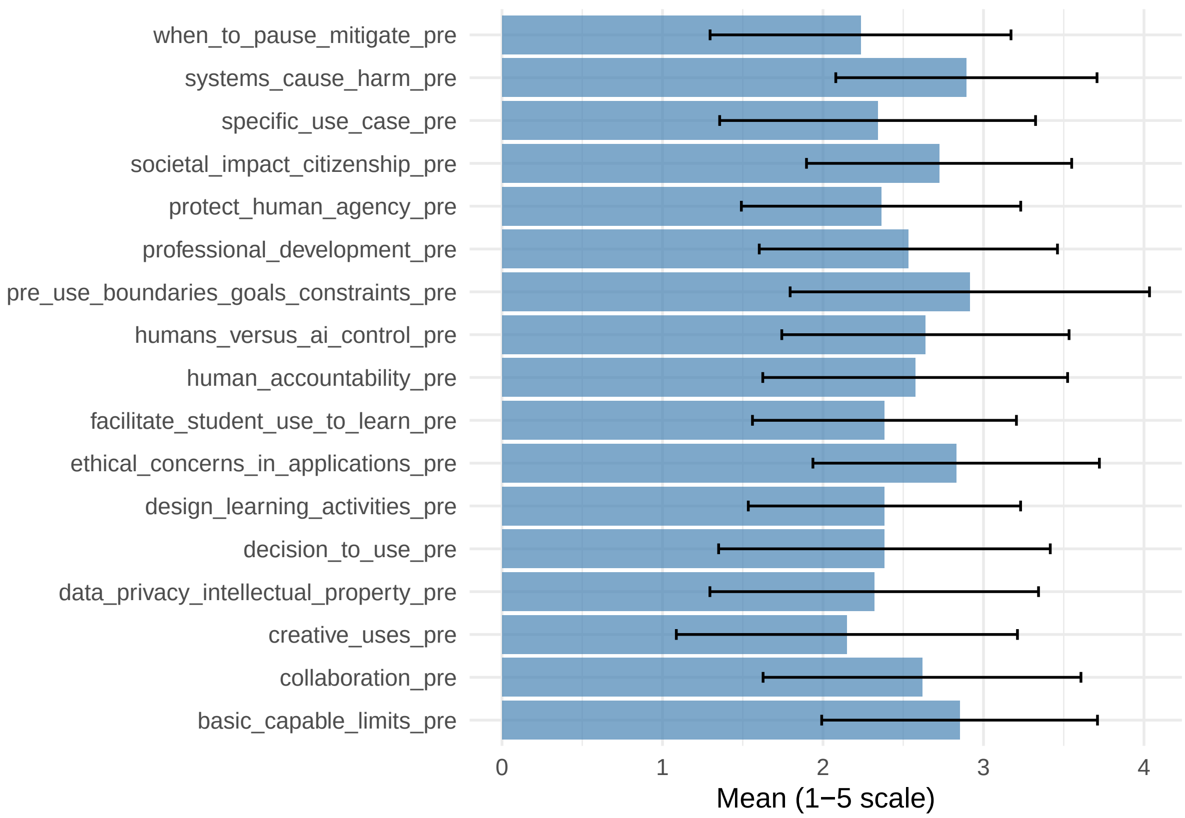


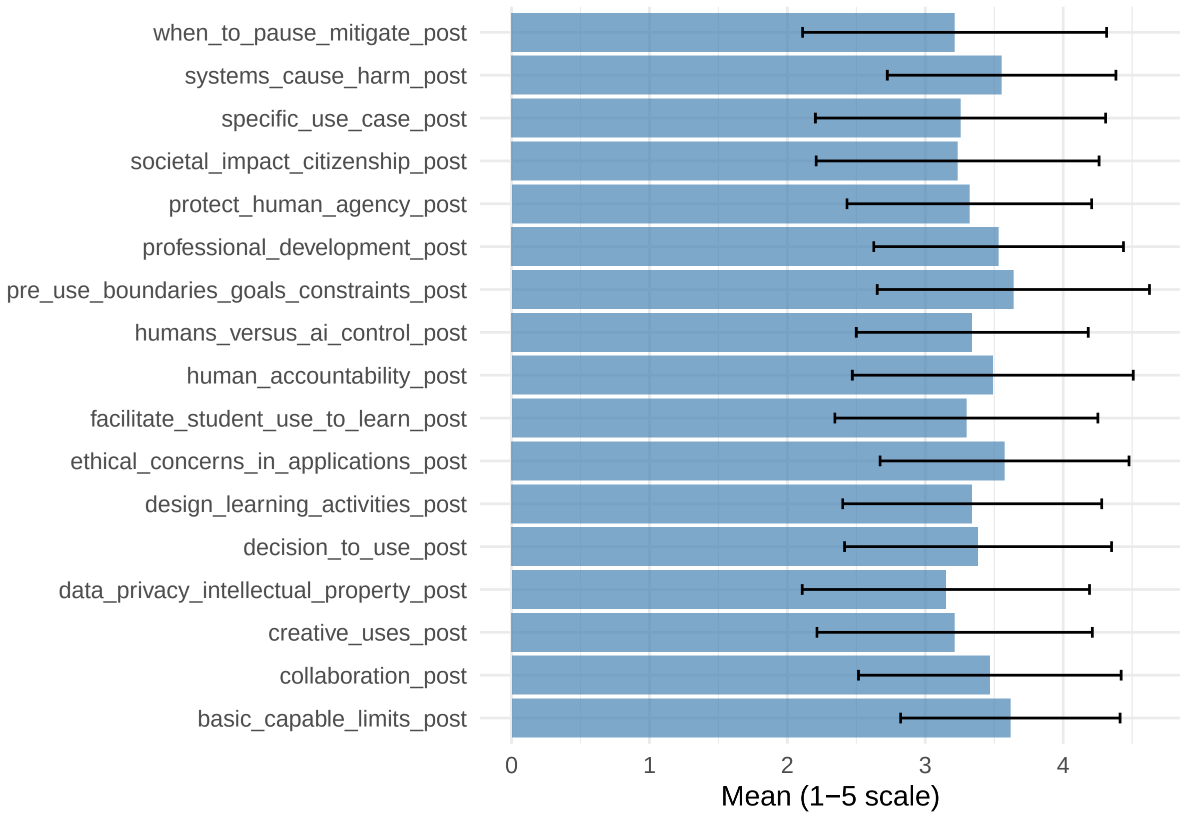
Figure 2.1: Pre-survey item means with +/-1 SD

Table 2.1 shows that most pre-survey means cluster between 2.2 – 2.9 on the 1-to-5 scale, so participants generally sit in the "Awareness-to-Acquire" band rather than at the extremes. As visualized in Figure 2.1, two items show the widest spread—`pre_use_boundaries_goals_constraints` ($\bar{X} \approx 2.9, s^2 \approx 1.25$) and `creative_uses` ($\bar{X} \approx 2.2, s^2 \approx 1.13$)—suggesting respondents differ markedly in how confidently they scope problems before using AI and in how creatively they apply tools. At the other end, items such as `systems_cause_harm` ($s^2 \approx 0.66$) and `societal_impact_citizenship` ($s^2 \approx 0.68$) display tighter clustering, indicating more consensus (or possibly limited discrimination) around basic harm-awareness themes.

No item shows a ceiling ($\bar{X} > 4$) or floor ($\bar{X} < 1.5$) effect, so all items retain headroom for change at the post-survey. The accompanying bar-and-error plot ($\bar{X} \pm 1SD$) visually confirms this mid-range pattern and flags which items have the most and least individual variability. That variability is helpful: it means there is enough dispersion to examine pre/post shifts and exploratory associations, although dispersion alone does not guarantee a stable factor solution.

Table 2.2: Item means and variances (Post-survey, N = 47)

item	mean	var
basic_capable_limits_post	3.62	0.63
collaboration_post	3.47	0.91
creative_uses_post	3.21	1.00
data_privacy_intellectual_property_post	3.15	1.09
decision_to_use_post	3.38	0.94
design_learning_activities_post	3.34	0.88
ethical_concerns_in_applications_post	3.57	0.81
facilitate_student_use_to_learn_post	3.30	0.91
human_accountability_post	3.49	1.04
humans_versus_ai_control_post	3.34	0.71
pre_use_boundaries_goals_constraints_post	3.64	0.98
professional_development_post	3.53	0.82
protect_human_agency_post	3.32	0.79
societal_impact_citizenship_post	3.23	1.05
specific_use_case_post	3.26	1.11
systems_cause_harm_post	3.55	0.69
when_to_pause_mitigate_post	3.21	1.21



Post-survey scores shift noticeably upward: most item means now cluster between $\bar{X} \approx 3.2$ and $\bar{X} \approx 3.6$, placing participants solidly in the "Acquire-to-Deepen" band (Table 2.2). The highest post-survey means appear on `pre_use_boundaries_goals_constr` ($\bar{X} \approx 3.64, s^2 \approx 0.98$) and `basic_capable_limits` ($\bar{X} \approx 3.62, s^2 \approx 0.63$) signalling broad confidence in scoping problems and understanding AI limits after the institute (Figure 2.2).

Variability remains highest for `when_to_pause_mitigate` ($s^2 \approx 1.21$) and `specific_use_case` ($s^2 \approx 1.11$), suggesting uneven mastery of risk-mitigation timing and domain-specific tool fit. The tightest clustering occurs on `systems_cause_harm` ($s^2 \approx 0.69$) and `basic_capable_limits` ($s^2 \approx 0.63$), indicating near-consensus on foundational harm awareness and tool capabilities. Importantly, no item approaches a ceiling effect ($\bar{X} < 4.0$), leaving room for additional growth. The consistently higher post-survey means also motivated the paired pre/post analyses reported later in this document.

2.3.1 Polychoric Correlation Matrix

Because all 17 survey items are ordered-categorical (1 … 5), estimating Pearson correlations on the raw codes would attenuate the true association among the underlying continuous traits. We therefore employ polychoric correlation, which posits a continuous, bivariate-normal latent pair (X, Y) that is divided into ordered categories by fixed cut-points.

Let X and Y be the latent variables with a bivariate normal distribution. The Latent formulation is:

$$(X, Y) \sim \mathcal{N}\left(\begin{bmatrix}0\\0\end{bmatrix}, \begin{bmatrix}1 & \rho\\ \rho & 1\end{bmatrix}\right)$$

The observed ordinal variables X, Y arise by slicing the latent axes at ordered thresholds

$$X = i \quad \Leftrightarrow \quad \tau_{i-1} < X^* \leq \tau_i, \quad Y = j \quad \Leftrightarrow \quad \gamma_{j-1} < Y^* \leq \gamma_j$$

with $\tau_0 = -\infty < \tau_1 < \tau_2 < \tau_3 < \tau_4 < \tau_5 = +\infty$ and $\gamma_0 = -\infty < \gamma_1 < \gamma_2 < \gamma_3 < \gamma_4 < \gamma_5 = +\infty$.

For each cell of the $k \times \ell$ contingency table we have

$$Pr(X = i, Y = j) = \int_{\tau_{i-1}}^{\tau_i} \int_{\gamma_{j-1}}^{\gamma_j} \phi(u, v; \rho)\, dv\, du$$

where $\phi(\cdot, \cdot; \rho) \;=\; \mathcal{N}\left(\begin{bmatrix}0\\0\end{bmatrix}, \begin{bmatrix}1 & \rho\\ \rho & 1\end{bmatrix}\right)$

The polychoric estimate $\hat{\rho}$ is the value of ρ that maximizes the likelihood of the observed cell counts.

The assumptions of the polychoric correlation are:

1. Underlying continuity & bivariate normality of X and Y.
2. Threshold invariance: the cut-points (e.g., between "Awareness" and "Acquire") are the same across all respondents.
3. Sufficient variability in category usage—otherwise thresholds or ρ become unstable.

Our descriptive step showed that item means were not concentrated at the scale endpoints and that the items retained useful variability (most $s^2 \in [0.66, 1.25]$). This is compatible with estimating thresholds, but it does not verify latent bivariate normality or threshold invariance.

The polychoric correlation coefficient is the maximum likelihood estimate of the correlation between two ordinal normal variables. The range of the polychoric correlation is from –1 to 1.

We inspect this matrix with a heatmap (Figure 2.3). Very high absolute correlations($|r| > .80$) may indicate true item duplication (*within-factor*) or may show an item sitting in the wrong competency (*cross-factor*) or may show an overarching wording effect.

The original scan returned 38 symmetric matrix entries, corresponding to 19 unique item pairs above the $|\hat{\rho}| > 0.80$ threshold. The unique pairs are listed in Table 2.3.

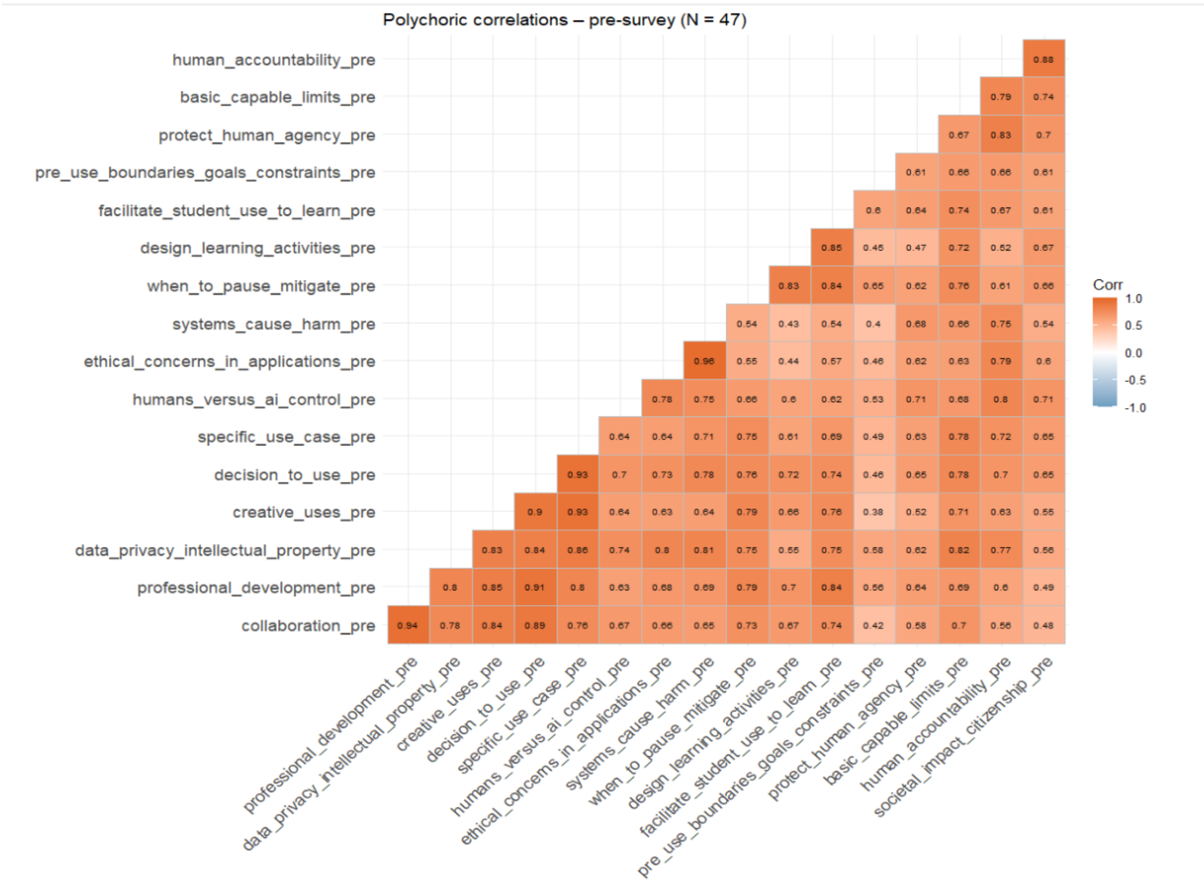


Figure 2.2: Polychoric correlations - pre-survey. Cells darker than 0.80 (absolute) are highlighted as potential redundancy candidates.

Table 2.3: Item pairs with |r| > 0.80 (pre-survey)

item1	item2	r
ethical_concerns_in_applications_pre	systems_cause_harm_pre	0.96
collaboration_pre	professional_development_pre	0.94
decision_to_use_pre	specific_use_case_pre	0.93
specific_use_case_pre	creative_uses_pre	0.93
decision_to_use_pre	professional_development_pre	0.91
decision_to_use_pre	creative_uses_pre	0.90
decision_to_use_pre	collaboration_pre	0.89
human_accountability_pre	societal_impact_citizenship_pre	0.88
data_privacy_intellectual_property_pre	specific_use_case_pre	0.86
design_learning_activities_pre	facilitate_student_use_to_learn_pre	0.85
creative_uses_pre	professional_development_pre	0.85
data_privacy_intellectual_property_pre	decision_to_use_pre	0.84
facilitate_student_use_to_learn_pre	when_to_pause_mitigate_pre	0.84
creative_uses_pre	collaboration_pre	0.84
facilitate_student_use_to_learn_pre	professional_development_pre	0.84
data_privacy_intellectual_property_pre	creative_uses_pre	0.83
protect_human_agency_pre	human_accountability_pre	0.83
design_learning_activities_pre	when_to_pause_mitigate_pre	0.83
data_privacy_intellectual_property_pre	basic_capable_limits_pre	0.82
systems_cause_harm_pre	data_privacy_intellectual_property_pre	0.81
ethical_concerns_in_applications_pre	data_privacy_intellectual_property_pre	0.80
data_privacy_intellectual_property_pre	professional_development_pre	0.80

In Table 2.4, I classify a small set of representative high-r pairs. The purpose is to distinguish likely wording redundancy from broader associations that may reflect a general AI-competence tendency.

Table 2.4: Representative high-correlation pairs and likely sources of overlap

Item1	Item2	Source
ethical concerns in applications	systems cause harm	Within Ethics; closely related harm judgments
professional development	collaboration	Within Professional Development; two-item domain overlap
specific use case	decision to use	Within Foundations; closely related tool-fit judgments
specific use case	creative uses	Within Foundations; selection and application overlap
human accountability	protect human agency	Within Human-Centered Mindset; accountability and agency overlap
facilitate student use to learn	design learning activities	Within Pedagogy; planning and facilitation overlap

The representative pairs show that the strongest overlap occurred within intended competency areas, especially Ethics, Foundations, Human-Centered Mindset, and Pedagogy. Cross-domain associations were also present, so high correlations could not be treated as automatic evidence that one item should be deleted.

We flag the following pairs for potential (drop vs.~parcel) action:

- `ethical_concerns` $\Longleftrightarrow$ `systems_cause_harm`
- `professional_development` $\Longleftrightarrow$ `collaboration`
- `specific_use_case` $\Longleftrightarrow$ `decision_to_use` and `creative_uses`

These flagged pairs feed directly into Exploratory Factor Analysis, where I examine how the pattern of shared variance relates to the trial pruning and bundling decisions.

2.3.2 Factorability

We use the Kaiser–Meyer–Olkin (KMO) measure of sampling adequacy and Bartlett's test of sphericity to assess whether our polychoric correlation matrix is suitable for factor analysis (Kaiser, 1974; Bartlett, 1950).

KMO summarizes how strongly the variables share common variance relative to their partial correlations. It is a diagnostic index rather than a null-hypothesis test. Values range from 0 to 1; values above 0.60 are often treated as minimally adequate for exploratory factor work.

Bartlett targets overall correlation strength. The null hypothesis for Bartlett's test is that the correlation matrix equals the identity ($R = I$) i.e., variables are uncorrelated in the population.

$$\chi^2 \; = \; -\Big[(n-1) - \tfrac{2p+5}{6}\Big] \; \ln|R|, \qquad \mathrm{df} = p(p-1)/2,$$

where n is the sample size, p is the number of variables and $|R|$ is the determinant of the correlation matrix. A significant result ($p < 0.05$) indicates that the correlation matrix is not an identity matrix, suggesting that factor analysis may be appropriate.

If $KMO \geq 0.60$ overall and Bartlett's test is significant ($p < 0.05$), we can cautiously attempt parallel analysis and EFA. Items with very low individual MSA (< 0.50) may be idiosyncratic. We will consider flagging them for review before the EFA run.

Table 2.5: KMO item-level MSA (overall = 0.678)

Item	MSA
humans_versus_ai_control_pre	0.60
protect_human_agency_pre	0.62
human_accountability_pre	0.69
societal_impact_citizenship_pre	0.68

ethical_concerns_in_applications_pre	0.77
systems_cause_harm_pre	0.76
data_privacy_intellectual_property_pre	0.66
basic_capable_limits_pre	0.65
decision_to_use_pre	0.83
specific_use_case_pre	0.73
creative_uses_pre	0.67
pre_use_boundaries_goals_constraints_pre	0.56
design_learning_activities_pre	0.59
facilitate_student_use_to_learn_pre	0.66
when_to_pause_mitigate_pre	0.64
collaboration_pre	0.68
professional_development_pre	0.75

Table 2.5 displays the KMO results. The overall KMO value of **0.678** indicates that the polychoric correlation matrix has "mediocre-middling' ' but is acceptable for factor analysis.

At the item-level MSAs, there are two items that sit just below the preferred 0.60 cutoff:

- `pre_use_boundaries_goals_constraints` (MSA = 0.56)
- `design_learning_activities` (MSA = 0.59)

Three more are right on the edge (≈ 0.60). These items share relatively little common variance with the rest of the matrix; they might cross-load weakly or act as noise. We will keep them for the initial EFA but watch for low communalities (< 0.20) or cross-loadings—both would justify dropping or parcelling them later.

Table 2.6: Bartlett test of sphericity

Statistic	Chi_sq	df	p_value
Chi-square	4223.08	136	<2e-16

Table 2.6 shows that the Bartlett's test is significant ($\chi^2(136) = 4223.08, p < 2e-16$). Since the null hypothesis $R = I$ is decisively rejected, we conclude that the correlation matrix contains systematic association. This finding supports attempting factor extraction, but it does not determine the number or stability of the factors.

Results Summary. The overall KMO was **0.678** (*middling*), while Bartlett's test was highly significant, $\chi^2(136) = 4223.08, \ p < 0.001$. Two items posted marginal individual MSAs—`pre_use_boundaries_goals_constraints` (0.56) and `design_learning_activities` (0.59)—suggesting they share limited common variance. I retain them for the first EFA and flag them for closer review if they show low communalities or substantial cross-loadings.

Taken together, the diagnostics support attempting dimensionality estimation via parallel analysis and MAP, with the small-sample caveat described above.

2.3.3 Determine Number of Factors

To investigate the latent dimensionality, two complementary, simulation-based criteria were applied to the 17 × 17 polychoric matrix:

1. Parallel Analysis (PA).

Horn's PA (1965) compares each empirical eigenvalue λ_i of R with the mean eigenvalue $\bar{\lambda}_i^{\text{rand}}$ obtained from 500 random data sets of size$n = 47,\ p = 17$, where the data drawn from a standard normal distribution with independent and identically distributed (iid) variables ($\mathcal{N}(\mathbf{0}, \mathbf{1})$). A factor is retained if

$$\lambda_i > \bar{\lambda}_i^{\text{rand}} .$$

2. Minimum Average Partial (MAP).

Velicer's MAP (1976) removes successive principal components, recomputing the average squared off-diagonal partial correlation,

$$\text{ASPC}_k = \frac{2}{p(p-1)} \sum_{i<j} r_{ij(k)}^2,$$

where $r_{ij(k)}$ is the partial correlation between variables i and j after extracting k components. The optimal dimensionality is the k that minimises ASPC_k.

Table 2.7: Factor-count suggestions (pre-survey)

Method	Suggested_k
Parallel analysis	—
MAP (minimum ASPC)	3

Table 2.7 shows the results of the PA and MAP tests. The polychoric matrix required light smoothing to achieve positive-definiteness, after which PA did not support the expected retained-factor solution (the first random eigenvalue exceeded the observed). MAP, however, exhibited a clear global minimum at $k = 3$. Ultra-Heywood warnings during high-factor trial solutions, together with the parallel-analysis result, indicate that the factor-count evidence is unstable at $N = 47$.

For diagnostic purposes, the exploratory factor analysis begins with three factors, with iterative trimming based on (i) communalities $h_i^2 < 0.20$ and (ii) primary loadings $|\lambda| < 0.32$. Because PA and MAP did not agree, the three-factor choice remains provisional.

2.3.4 Exploratory Factor Analysis (EFA)

In an exploratory-factor-analysis (EFA) of Jöreskog 1969, the $p = 17$ vector of standardized item scores $\mathbf{x} \in \mathbb{R}^p$ can be written as a linear measurement model

$$\mathbf{x} = \Lambda\mathbf{f} + \varepsilon, \quad \text{with } \mathbf{f} \sim \mathcal{N}(\mathbf{0}, \Phi),\ \varepsilon \sim \mathcal{N}(\mathbf{0}, \Psi),$$

where Λ is the $p \times k$ factor loading matrix, $\mathbf{f}$ is the k-dimensional latent factor vector, Φ is the factor covariance matrix, and Ψ is $diag(\Psi_1, \dots, \Psi_p)$, containing unique variance (error) covariance matrix.

The EFA represents the observed polychoric correlation matrix R with a k-factor model and zero off-diagonal elements in Ψ:

$$\mathbf{R} \approx \Lambda\Phi\Lambda^\top + \Psi.$$

Assumptions are: (i) moderate multivariate normality of the factors, (ii) uncorrelated uniques, and (iii) correct rank k (Horn, 1965; Velicer, 1976).

Extraction proceeded with principal-axis factoring applied to the smoothed polychoric correlation matrix, a method recommended for non-normal or ordinal indicators because it relies on the communalities rather than on the total variances of the items (Gorsuch, 1983). Dimensionality was fixed provisionally at $k = 3$ on the basis of Velicer's minimum-average-partial (MAP) criterion; parallel analysis did not yield an interpretable retained-factor recommendation after the correlation-matrix instability and smoothing (Horn, 1965). The resulting factors were rotated with oblique Promax rotation ($k = 4$), which allows correlated factors (Jennrich and Sampson, 1966).

The diagnostics supported attempting exploratory factor modelling, with caution. The overall Kaiser–Meyer–Olkin index was $KMO = 0.678$, classed as "middling" (Kaiser, 1974). Bartlett's test of sphericity was highly significant, $\chi^2(136) = 4223.1, p < 0.001$, rejecting the null hypothesis that $R = I$ and indicating systematic covariance among the items. After extraction, every communality exceeded $h_i^2 = 0.20$ (MacCallum et al., 1999). No item failed that diagnostic in this solution, but the result did not establish stable item retention.

Table 2.8: Pattern loadings ≥ 0.32 (promax, 3-factor)

Item	PA1	PA3	PA2
humans_versus_ai_control_pre	0.10	0.43	0.44
protect_human_agency_pre	0.00	0.60	0.32
human_accountability_pre	-0.20	0.79	0.47
societal_impact_citizenship_pre	-0.05	0.89	0.07
ethical_concerns_in_applications_pre	0.01	0.14	0.84
systems_cause_harm_pre	0.05	0.02	0.92
data_privacy_intellectual_property_pre	0.48	0.08	0.48
basic_capable_limits_pre	0.38	0.50	0.12
decision_to_use_pre	0.76	-0.06	0.34
specific_use_case_pre	0.64	0.05	0.30

creative_uses_pre	0.89	-0.15	0.20
pre_use_boundaries_goals_constraints_pre	0.07	0.70	-0.03
design_learning_activities_pre	0.76	0.38	-0.33
facilitate_student_use_to_learn_pre	0.74	0.35	-0.16
when_to_pause_mitigate_pre	0.74	0.38	-0.18
collaboration_pre	0.89	-0.18	0.23
professional_development_pre	0.90	-0.11	0.18

Table 2.9: Item communalities (h^2)

Item	h2
humans_versus_ai_control_pre	0.72
protect_human_agency_pre	0.68
human_accountability_pre	0.97
societal_impact_citizenship_pre	0.80
ethical_concerns_in_applications_pre	0.87
systems_cause_harm_pre	0.93
data_privacy_intellectual_property_pre	0.85
basic_capable_limits_pre	0.78
decision_to_use_pre	0.93
specific_use_case_pre	0.81
creative_uses_pre	0.87
pre_use_boundaries_goals_constraints_pre	0.54
design_learning_activities_pre	0.76
facilitate_student_use_to_learn_pre	0.83
when_to_pause_mitigate_pre	0.85
collaboration_pre	0.85
professional_development_pre	0.90

The polychoric matrix required smoothing and the trial solutions produced temporary Heywood warnings. Later trimming reduced those warnings, but it did not remove the underlying small-sample limitation.

Table 2.8 shows how strongly each survey item "loads" onto each of the three extracted factors. A loading is a correlation-like value between an item and a factor. PA1, PA3, and PA2 are software labels that follow extraction order, not theoretical importance. Values near 0.30 or higher were treated as meaningful for this exploratory review. For example, `creative_uses` loaded 0.89 on PA1 and lower on PA3 and PA2. Other items loaded on more than one factor, suggesting overlap between specific competency areas and a broader AI-competence tendency.

Table 2.9 shows item communalities (h^2), the proportion of each item's variance reproduced by this factor solution. The estimates were all at least 0.54, but high communalities can reflect strong shared variance and redundancy as well as useful common structure. I therefore did not interpret them as proof that every item should be retained.

Results. The three-factor promax solution described broad clusters related to tool use and professional growth, human agency and pedagogy, and ethics and harm. Several items cross-loaded, consistent with a general AI-competence tendency that cut across the intended domains. Because the polychoric matrix required smoothing, parallel analysis was inadmissible, and $N = 47$ was small relative to the number of items, I treat this solution as exploratory. It helped identify redundancy and possible structure, but it did not validate the instrument or confirm a stable three-factor model.

2.3.5 Bifactor / Hierarchical Model

In a bifactor structure, every observed, standardized item (x_i) loads on one general factor G that captures shared "AI competence," plus one orthogonal group factor ($S_{j(i)}$) that represents its specific competency domain; residuals remain uncorrelated:

$$x_1 = \lambda_{Gi} G + \lambda_{S_{j(i)} i} S_{j(i)} + \varepsilon_i \quad \text{with } Cov(G, S_j) = 0 \quad (j \neq j').$$

In matrix form

$$\mathbf{R} = \Lambda_G \Lambda_G^\top + \Lambda_S \Lambda_S^\top + \Psi.$$

Redundancy is flagged when an item retains large loadings on both G and its group factor (here we use $|\lambda| \geq 0.30$). Items whose group-specific loading shrinks toward zero after G is extracted add little unique information in that solution and become candidates for closer content review.

Reliability attribute to the general factor is quantified by McDonald's hierarchical ω_h (McDonald, 1999; Reise, 2012).

$$\omega_h = \frac{(\mathbf{1}^\top \Lambda_G)^2}{\mathbf{1}^\top \mathbf{R} \mathbf{1}},$$

where the general factor is standardized, Λ_G is the vector of general-factor loadings, and the denominator is the variance of the unit-weighted total score.

Larger values of ω_h suggest that a general factor contributes more strongly to total-score variance, while smaller values leave more room for domain-specific variance. I later attempted confirmatory and invariance models, but those models were not stable enough to support confirmatory conclusions.

Key assumptions are (i) orthogonality between the general and group factors, (ii) uncorrelated uniquenesses ($Cov(\epsilon_i, \epsilon_j) = 0$), and (iii) a correct choice of factor count inherited from the PA/MAP step.

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## [1] 0.82
```

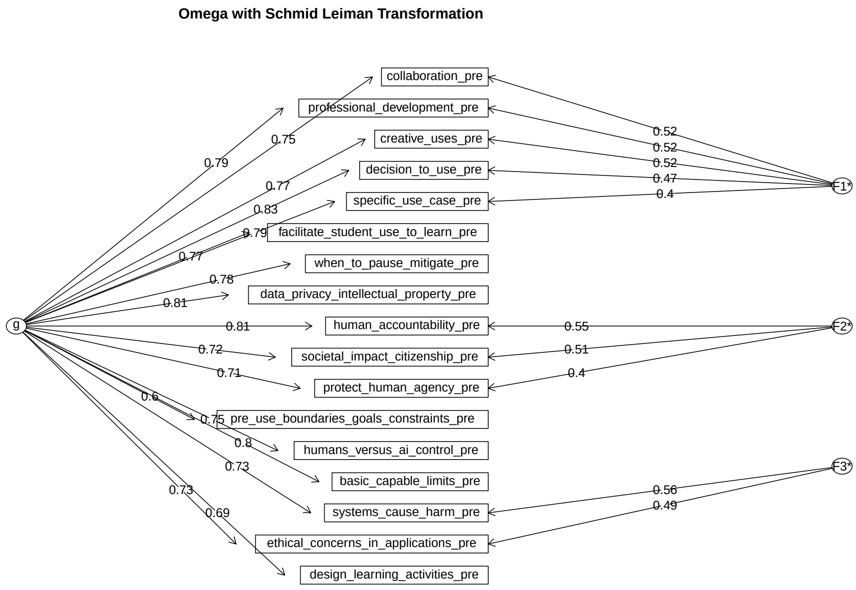

Figure 2.3: Exploratory bifactor representation for the 17 pre-survey items. The diagram records the attempted general and group-factor solution; the estimates are provisional because of the small sample, matrix smoothing, and earlier factor-extraction warnings.

The exploratory bifactor calculation produced $\omega_h \approx 0.82$, suggesting a strong general component in this sample. The group factors also retained domain-specific variation. However, the same small-sample and matrix-stability concerns that affected the EFA apply here. I therefore use the bifactor result as a diagnostic for shared variance and item overlap, not as confirmation of a general-factor model or as a basis for claiming that the instrument was validated.

2.3.6 Reliability & Redundancy Checks

The exploratory bifactor result suggested a strong general component, but reliability must be interpreted separately from dimensionality. I therefore estimated internal consistency and average inter-item association while also looking for evidence of redundancy.

Cronbach's α summarizes the internal consistency of a set of items under assumptions about their measurement properties. A high value can indicate coherence, but it does not test a null hypothesis, prove unidimensionality, or establish validity.

$$\alpha = \frac{k}{k-1}\left(1 - \frac{\sum_{i=1}^{k} \sigma_i^2}{\sigma_{total}^2}\right),$$

where k is the number of items, σ_i^2 is the variance of item i, and σ_{total}^2 is the variance of the total score. The coefficient is interpreted here as an exploratory reliability diagnostic (Cronbach, 1951), alongside ω, average inter-item correlation, and item content.

McDonald's ω_t reflects total reliability under a common factor model when loadings are unequal.

$$\omega_t = \frac{\mathbf{1}^\top \Lambda \Phi \Lambda^\top \mathbf{1}}{\mathbf{1}^\top \mathbf{R} \mathbf{1}},$$

where Λ is the loading matrix, Φ is the factor covariance matrix, and the denominator is the total-score variance (McDonald, 1999).

Average inter-item correlation quantifies the average correlation between all pairs of items.

$$\bar{r} = \frac{2}{k(k-1)} \sum_{i<j} r_{ij}$$

Following (Briggs and Cheek, 1986), the ideal range is $0.15 \leq \bar{r} \leq 0.50$. Values > 0.50, together with very high α/ω, flag potential duplication.

Finally, the "α-if-deleted'' table highlights items whose removal boosts α by more than 0.02 (Nunnally and Bernstein, 1994). Items that also make little unique contribution in the exploratory bifactor loadings are candidates for closer content review. Any pruning or parcelling remains provisional rather than a confirmatory decision.

Table 2.10: Cronbach's α-if-deleted (pre-survey)

Item	std.alpha	G6(smc)	average_r
humans versus ai control pre	0.96	0.98	0.63
protect human agency pre	0.97	0.98	0.63
human accountability pre	0.96	0.98	0.63
societal impact citizenship pre	0.97	0.98	0.63
ethical concerns in applications pre	0.97	0.98	0.63
systems cause harm pre	0.97	0.98	0.63
data privacy intellectual property pre	0.96	0.98	0.62
basic capable limits pre	0.96	0.98	0.63
decision to use pre	0.96	0.98	0.62
specific use case pre	0.96	0.98	0.62
creative uses pre	0.96	0.98	0.63
pre use boundaries goals constraints pre	0.97	0.99	0.65
design learning activities pre	0.97	0.98	0.64
facilitate student use to learn pre	0.96	0.98	0.63

when to pause mitigate pre	0.96	0.98	0.63
collaboration pre	0.96	0.98	0.63
professional development pre	0.96	0.98	0.62

Table 2.11: Overall reliability statistics (pre-survey)

Statistic	Value
Cronbach α (std.)	0.97
Average inter-item r	0.63
McDonald ω_total	0.98
McDonald ω_hierarchical	0.82

Table 2.10 shows that no single item materially lowers the full-set alpha. Table 2.11 shows very high internal-consistency estimates ($\alpha = 0.97$, $\omega_t = 0.98$) and an average inter-item correlation of 0.63. Rather than treating those values as proof of scale quality, I interpret them as evidence that many items share substantial information. The provisional $\omega_h = 0.82$ is consistent with a strong general component, but its precision is limited by the sample and exploratory factor solution.

2.3.7 Exploratory Iterative Pruning / Bundling

This trial examined what happened when highly overlapping items were removed or bundled while preserving as much conceptual coverage as possible. After each pass, I recomputed α, ω, and average inter-item correlation. The exercise was diagnostic: statistical overlap alone was not sufficient to make a final item-retention decision.

Pass 0 (17 original items) exhibited very high internal consistency (Cronbach's $\alpha = 0.97$; $\bar{r} = 0.64$). During Pass 1, I averaged `decision_to_use`, `specific_use_case`, and `creative_uses` into a trial "tool-use" parcel and removed one item from several highly associated pairs. These choices recorded one possible reduction route; they did not establish that the removed content was unnecessary.

After the first trim, $\alpha = 0.95$ and $\bar{r} = .64$, so the retained item set still contained substantial overlap. The estimated ω_h fell to .68, but the instability of the exploratory factor solutions limits substantive interpretation of that change.

The second pass addressed additional very highly correlated items by removing `data_privacy_intellectual_property`, `when_to_pause_mitigate`, and `humans_versus_ai_control`. Although the pairwise associations motivated the trial, these removals also reduced the breadth of content represented by the item set.

After the second trim and with nine items, $\alpha = 0.93$ and $\bar{r} = 0.63$, which remains above the 0.15–0.50 reference range. The estimated ω_h was 0.77; because it came from the same unstable exploratory structure, I do not treat the threshold as proof of a dominant general factor (Reise, 2012).

Pass 3 produced a trial reduced scoring set. I parcelled `protect_human_agency` and `human_accountability` into an "agency–accountability" score; retained `societal_impact_citizenship`, `systems_cause_harm`, the tool-use parcel, and `design_learning_activities`; and removed `basic_capable_limits` and `professional_development`. This reduced overlap but left some competency areas represented by a single item and omitted the Professional Development area from the trial set.

The third pass tells us that $\alpha = 0.89$ and $\bar{r} = 0.62$ which is still high given that only five manifest scores remain. $\omega_h = 0.38 (< .70)$, suggesting that a general factor accounted for less of the reliable total-score variance in this trial solution. At this point, further deletion would have under-represented at least one competency area. I stopped the trial and treated the retained scores as a diagnostic illustration rather than a finalized instrument or scoring plan.

The trial reduction suggested three broad clusters (Human, Tool/Pedagogy, and Ethics), but the result is sample-dependent and does not establish a confirmed three-domain structure.

- The general-factor estimate is weaker in this reduced set, and the bifactor solution remains unstable.
- For descriptive review, the five retained scores can be grouped into three broad content clusters:

1. Human-agency parcel (agency + accountability)
2. Human-societal impact single item
3. Tool-use parcel
4. Design-learning single item
5. Systems-cause-harm single item

Tool & Pedagogy still correlate, but content is distinct.

Iterative pruning reduced internal consistency from $\alpha = 0.97 \rightarrow 0.89$ and $\omega_{\text{total}} = 0.98 \rightarrow 0.91$; meanwhile the hierarchical coefficient fell to $\omega_h = 0.38$ (McDonald, 1999). These changes show that the reliability estimates were sensitive to which items and parcels were retained. They do not, by themselves, establish a three-domain scoring structure.

1. Human-Centered Mindset: agency_account_parcel (agency + accountability) and `societal_impact_citizenship` (single item)
2. Tool Foundations & Pedagogy: tool_use_parcel (tool selection & adaptation) and `design_learning_activities` (single item)
3. Ethics of AI: `systems_cause_harm` (single item)

Table 2.12: Means and standard deviations of retained scores

Score	M	SD
Societal impact	2.72	0.83
Systems harm	2.89	0.81
Tool use	2.29	0.98
Agency / accountability	2.47	0.85
Learning activities	2.38	0.85

Table 2.13: Inter-score correlations

Score	Societal	Systems	Tool	Agency	Learning
Societal impact	1.00	0.50	0.65	0.78	0.62
Systems harm	0.50	1.00	0.67	0.67	0.41
Tool use	0.65	0.67	1.00	0.68	0.70
Agency / accountability	0.78	0.67	0.68	1.00	0.48
Learning activities	0.62	0.41	0.70	0.48	1.00

Descriptive statistics for the retained scores (Table 2.12) show mid-scale means ($2.3 \leq M \leq 2.9$) and adequate spread ($SD \approx 0.8$–1.0), leaving head-room for post-survey change. Inter-score correlations (Table 2.13) remain moderate to high ($r = 0.41$–0.78), signalling related but distinguishable constructs; the largest association joins the two human-centered indicators ($r = 0.78$). The within-parcel alphas were 0.83 for agency–accountability and 0.88 for tool use. These values describe consistency within the trial parcels; they do not establish that bundling preserved all relevant information. The findings document what the pruning exercise retained and what conceptual coverage it lost. For the public pre / post summary, I return to the five competency areas defined in the final survey instruments.

2.3.8 Attempted Confirmatory and Invariance Models

I also attempted confirmatory factor and pre / post invariance models. The latent CFA solutions were inadmissible or unstable at $N = 47$. A later saturated manifest model had zero degrees of freedom, so its perfect fit reproduced the observed matrix by definition and did not test whether the proposed structure was correct. The invariance attempt also treated the two waves as groups even though the same respondents completed both surveys. I therefore do not use those models to claim confirmed factor structure, longitudinal invariance, or validated scores. The attempts remain part of the analytical record because they explain why I moved to a simpler paired summary.

2.4 Pre/Post Change in the Five Intended Competency Areas

The psychometric work showed meaningful item overlap but did not establish a definitive reduced instrument. For the public pre / post summary, I therefore returned to the five competency areas and all 17 items specified in the final Qualtrics instruments. Within each area, I averaged the item ratings for each respondent at each wave.

I used paired Wilcoxon signed-rank tests because the same 47 respondents completed both surveys and the scores are based on ordered response categories. Positive Hodges-Lehmann estimates indicate higher post-survey ratings. I report matched-pairs rank-biserial correlations and use Holm's adjustment across the five comparisons.

Table 2.14: Matched pre/post comparisons for the five intended competency areas

Competency	Pre M (SD)	Post M (SD)	HL shift [95% CI]	Rank-biserial r	Holm p
Human-Centered Mindset	2.57 (0.79)	3.35 (0.87)	0.75 [0.62, 1.00]	0.93	$< .001$
Ethics of AI	2.68 (0.84)	3.43 (0.86)	0.67 [0.50, 1.00]	0.90	$< .001$
AI Foundations and Applications	2.43 (0.91)	3.37 (0.89)	1.00 [0.75, 1.25]	0.94	$< .001$
AI Pedagogy	2.48 (0.78)	3.37 (0.84)	0.88 [0.63, 1.12]	0.92	$< .001$
AI for Professional Development	2.57 (0.93)	3.50 (0.89)	1.00 [0.50, 1.25]	0.90	$< .001$

All five competency-area summaries were higher at the post-survey. The estimated paired shifts were approximately three-quarters to one response category, and the nonzero changes were overwhelmingly in the upward direction. These results describe a clear pattern in the participants' self-ratings. They do not isolate the Institute's causal contribution or establish that the survey measures exactly the same construct in exactly the same way at both waves.

2.5 What the Analysis Supported

The available evidence supported several narrower conclusions:

- The final pre- and post-surveys used the same 17 competency statements across five intended areas.
- The items shared substantial variance, with several pairs showing likely content redundancy.
- Exploratory factor methods were useful for diagnosing overlap, but the sample did not support a confirmed factor structure.
- The trial pruning exercise illustrated how redundancy and content coverage compete when developing a shorter score.
- Aggregate self-ratings were consistently higher at the post-survey among the 47 matched respondents.

The evidence did not establish a validated instrument, a definitive reduced scoring system, longitudinal measurement invariance, or a causal effect of the Institute.

2.6 Limitations

This was a proof-of-concept evaluation with a small, self-selected sample from one faculty development program. The competency responses were self-ratings rather than direct demonstrations of performance. Sparse ordinal cells and the ratio of respondents to items limited factor estimation. There was no comparison group or random assignment, and the paired pre/post design cannot rule out response-shift effects, repeated-measurement effects, or other events occurring during the same period.

2.7 Analytical Reflection

This analysis began as a reliability check before comparing pre- and post-survey scores. It became a broader investigation of how item redundancy, factor structure, sample size, and scoring decisions affect what can be claimed from a new instrument. The unsuccessful factor and invariance routes were important because they showed where the data could not support the model I was asking it to estimate.

The most useful decision was to narrow the conclusion. I used the exploratory evidence to document the measurement concerns and returned to a transparent summary based on the five areas defined in the final surveys.

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